tube.xyz: an interactive structural atlas of the tubulin code

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Motivation

The *tubulin code* — combinations of α/β isotypes and post-translational modifications (PTMs) — modulates microtubule composition and the interactions of tubulin with motors, MAPs and small molecules. The PDB now holds ~1000 tubulin-containing structures and curated compendia such as the TubulinDB (Abbaali *et al.* 2023) catalogue thousands of disease-linked mutations and PTMs — but these structural and annotation layers live in separate resources and resist co-exploration.

tube.xyz places every tubulin residue from every deposited structure into a single *coordinate system* per family (α, β, γ, δ, ε). Sequence variants, ligand-binding contacts, literature mutations and PTMs are programmatically lifted onto the common residue index. The substrate is deliberately **pluggable**: new sources — clinical variants, fragment screens, new structures — join via a simple mapping with no schema changes. A 2D MSA coupled to a Mol*-based 3D viewer plus an LLM-driven chatbot facilitate navigation.

Database overview

Layer	Source	Count
Structures	RCSB PDB (IPR000217)	~1000
Tubulin families	HMM ($\alpha, \beta, \gamma, \delta, \varepsilon$)	5
MAP / enzyme families	HMM (kinesins, EBs, TTLLs, ...)	39
α/β isotypes	UniProt (TUBA1A–TUBB8)	16
Lit. mutations	TubulinDB (Abbaali 2023)	~5,200
Lit. modifications	TubulinDB + Janke 2020	~1,950
Binding sites	Mol* → master coords	per-structure
Taxonomy	NCBI / ete3 lineages	hundreds

Pipeline — every PDB structure is HMM-classified into a tubulin or MAP family, isotype-called against the human reference set, MUSCLE-aligned to a per-family master MSA, and persisted in Neo4j. Literature variants and PTMs from TubulinDB are mapped onto the same residue index via Universal Tubulin Numbering. New annotation sources plug in as soon as they carry a residue position.

Showcase: TUBB3 disease mutations on the structured face of β-tubulin

figures/showcase_3d.png
Cartoon of TUBB3 (chain B of 6S8L, 1.80 Å X-ray),
5 disease-mutation positions coloured by phenotype,
GTP & plinabulin shown for context.

• CFEOM (eye) • cortical dysplasia / MCD • biochemical (in vitro)

R262 6S8L:B · master 262

Substitutions reported
R → C • CFEOM
R → H • CFEOM
R → A • kinesin defect

Conservation R conserved in all 9 human β-isotypes.

Ref. Tischfield *et al.* (2010) *Cell.*

figures/showcase_msa.png
Nightingale MSA: TUBB1, TUBB2A, TUBB2B, **TUBB3**, TUBB4A, TUBB4B, TUBB5, TUBB6, TUBB8 (top to bottom),
master positions ~200–420 visible, mutation columns 205 / 262 / 401 / 410 / 417 boxed.

All five TUBB3 disease positions above are identical across every human β-isotype — pathogenic precisely because they sit at residues the protein cannot afford to vary. The aux row above the MSA shows literature-variant density per column from TubulinDB.

Other queries the chatbot handles

#1 Filter + binding site	*"Where does taxol bind in a human tubulin structure resolved at or below 3.5 Å?"* — faceted catalogue of human β-tubulin with TA1/TXL bound, sorted by resolution; taxane-site contact residues lifted onto the master alignment.
#2 Cross-species pocket comparison	*"Compare the GTP binding site in human and* Toxoplasma gondii *α-tubulin."* — resolves to `6S8L:A` (human, 1.80 Å) and `9Y9Z:1A` (*T. gondii*, 3.14 Å); 25 of ~32 GTP-contact positions shared in master coords.
#3 Annotation transfer	*"I'm interested in phosphorylation of* T. gondii *tubulin, but there is data only for human — compare sequences and highlight phosphorylation sites."* — aligns the *T. gondii* chain to the human master, then overlays human-derived phosphorylation as a PTM track on the alignment.

Key refs. [1] Abbaali *et al.* (2023) *PLoS ONE* **18**: e0295279 (TubulinDB). [2] Janke & Magiera (2020) *Nat. Rev. Mol. Cell Biol.* **21**, 307. [3] Steinmetz & Prota (2018) *Trends Cell Biol.* **28**, 776.